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Birth of the viru microplate triggered by Pacific-Farallon plate reorganization

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ABSTRACT

Chaotic spreading episodes create complex seafloor features, such as oblique tectonic fabric and pseudofaults, that challenge the assumptions of classic plate tectonic theory. Such episodes are often linked to major plate reorganizations, yet their detailed evolution and implications remain poorly constrained. Here we present geophysical evidence for a previously undocumented chaotic spreading event (44–28 Ma) and the formation of a new microplate, the Viru Microplate, at the Pacific-Farallon plate boundary in the eastern Pacific. By integrating high-resolution gravity data from the recent Surface Water Ocean Topography (SWOT) mission with new multibeam bathymetry collected during a 2024 R/V *Kilo Moana* voyage, we resolve a sequence of ridge propagation events, ridge jumps, and microplate formation that accompanied this chaotic episode. The spatial and temporal synchronism of these features with major reorganizations of the Pacific and Farallon plates suggests that global-scale plate stress field changes directly modulated divergent boundary behavior. Our findings demonstrate that global plate reorganizations can drive local-scale ridge complexity and microplate formation far from the triggering sites, providing diagnostic markers for regional and global tectonic change.

1. Introduction

Regions of oceanic crust containing seafloor features inconsistent with simple seafloor spreading models (Vine and Matthews, 1963), such as microplates, pseudofaults, and oblique tectonic fabric, may coincide with major plate reorganizations (Engeln et al., 1988). Dating these features can provide constraints on the timing of such reorganizations. For example, Matthews et al. (2016) determined that the formation of the Mammerickx microplate coincided with the India–Eurasia collision, suggesting that knowledge of the age of microplate formation could refine constraints on the timing of the collision. Others have arrived at similar conclusions, identifying doubled magnetic anomalies characteristic of microplate formation that were created during major plate boundary reorganizations (Menard, 1978; Mammerickx et al., 1980).

The seafloor created during plate reorganizations and microplate formation is typically characterized by confusing sequences of features resulting from ridge propagation and ridge jumping processes (Hey, 1977; Small, 1995). These processes create areas of asymmetric accretion at the formative spreading ridges (Müller et al., 1998), especially where spreading regimes are characterized as fast. Generally, spreading asymmetry appears to increase with spreading rate except at ultraslow spreading regimes (Seton et al., 2020). Additionally, most microplates are found in the Pacific Ocean (Matthews et al., 2016) along the East Pacific Rise, where spreading rates and asymmetric accretion are among the highest in the world (Seton et al., 2020). Although asymmetric spreading also occurs at ultra-slow spreading regimes, such as the Southwest Indian Ridge area, there are no documented examples of microplates in those regions (Matthews et al., 2016). Ridge propagation and jumping are driven either by lithospheric instability as a result of nearby hotspot magmatism within 1000 km (Schilling, 1985, 1991; Small, 1995; Ribe and Delattre, 1998; Müller et al., 1998; Mittelstaedt et al., 2008) or by changes in plate motion that alter the stress regime at spreading centers (Schouten et al., 1993; Bird and Naar, 1994; Eakins and Lonsdale, 2003; Hey, 2004). It has also been suggested that ridge propagation occurs down topographic gradients (Morgan and Sandwell, 1994).

Ridge propagation is a process by which a mid-ocean ridge spreading segment continuously "jumps" in the direction of propagation into crust of a contrasting age, leaving behind a series of segments of the abandoned ridges (Hey, 1977). Evidence of ridge propagations include propagation wakes along which younger crust lies adjacent to older crust, separated by a specific type of transform fault referred to as a

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pseudofault (Hey, 1977). Additionally, the abandoned spreading centers may appear as a series of deep troughs, each positioned further south or north than the previous depending on the direction of spreading at the ridge segment (Mammerickx et al., 1988). Ridge jumping occurs when a "dying" spreading center is abandoned in favor of a new one (Hey, 1977; Small, 1995). They are identified by fossil spreading ridges left behind after the cessation of spreading during a ridge jump (Mammerickx et al., 1980) and can occur on shorter and longer spatial scales (Small, 1995). During ridge propagation, the ridge jumps occur on much smaller scales (Hey, 1977), compared with ridge jumping that occurs when entire ridge segments are abandoned (Mammerickx and Sandwell, 1986).

Microplates, while active, are defined as independently rotating sections of crust bounded by two spreading ridges (Mammerickx and Klitgord, 1982; Mammerickx et al., 1988). Dual spreading and propagation occur at the bounding spreading ridges, accreting crust between the ridges and creating the microplate core. Eventually, spreading ceases at one of the bounding ridges, transferring the microplate to one of the adjacent major plates, after which the microplate is considered inactive and sometimes referred to as a paleoplate (Mammerickx et al., 1988). Spreading continues at the remaining ridge, and the microplate is physically defined as extending from the failed fossil ridge to a boundary where full spreading restarted (Mammerickx et al., 1988). Evidence of microplates includes failed spreading ridges, abyssal hills with varying characteristics, pseudofaults, and residual features on the conjugate flank of the spreading ridge (Mammerickx et al., 1988).

The chaotic seafloor formed due to the ~50 Ma Pacific plate reorganization at its southern boundaries with the Nazca, Cocos, and Antarctic plates is mostly found in remote regions of the ocean. The global coverage of gravity data derived from satellite altimetry is useful for studying these regions but lacks the resolution necessary for mapping smaller structures that result from reorganizations. High resolution multibeam bathymetry is the best method for mapping the seafloor in fine detail, however only about 27% of the world's oceans have been mapped in this way (Seabed 2030 Project, 2025). The newly launched SWOT satellite mission (Fu et al., 2024) improves upon both the resolution of past satellite altimeters as well as the coverage of multibeam bathymetry. It is therefore used in conjunction with bathymetry data to provide full coverage of the study area.

We present new swath bathymetry data collected on a research voyage aboard the R/V *Kilo Moana* in 2024 to a remote region of the southeast Pacific Ocean formed during the Paleogene between the Galapagos and Marquesas fracture zones, about 800 km east of the Marquesas Islands (Fig. 1). Together with the SWOT VGG, this data revealed adjacent abyssal hills sitting at different orientations to each other within one ridge segment, which is inconsistent with simple conveyor belt seafloor spreading where abyssal hills in a single ridge segment should form parallel to the ridge (Vine and Matthews, 1963). Further study of the surrounding area led to the identification of notable features in the region, interpreted as a series of ridge propagation and ridge jumping events, as well as the formation of a previously undiscovered microplate. We have named this microplate the "Viru" microplate, for the nearby Viru West Fracture Zone. Other features in the spreading corridor include additional microplates, various spreading centers, and the Peru-Chile subduction zone. In the corresponding region of the Nazca plate lie the Mendaña and Viru fracture zones conjugate to the Marquesas and Viru West fracture zones in the study area.

The remote nature of this region and proximity to the magnetic equator has resulted in a lack of magnetic anomaly data in the study area. The limited available information indicates that the first of the observed features formed around C20n, corresponding to 43.789 Ma on the Gee and Kent (2007) timescale, while magnetic anomaly picks for C12n (30.939 and 30.479 Ma), C11n.2n (30.098 Ma), C11n.1n (29.662 and 29.401 Ma), and C10n.2n (28.745 Ma) coincide with the microplate, indicating its formation around this time. The timing of these features as well as the consideration of tectonic events at the time lead to the conclusion that these features occurred during a period of complex spreading that may have been triggered by the major plate reorganization that led to a change in the Pacific plate direction around 52–45 Ma (Gaastra et al., 2022; Torsvik et al., 2017; Barckhausen et al., 2013; Menard and Atwater, 1968), while the capture of the microplate by the Pacific plate coincides with the breakup of the Farallon plate starting around 28 Ma (Atwater and Stock, 1998; Lonsdale, 2005).

2. Methods

2.1. Mapping and dating

The mapping data used were the calculated vertical gravity gradient (VGG) from Yu et al. (2024), of SWOT's level 2 data (Fu et al., 2024), collected from July 2023 to July 2025, as well as swath bathymetry collected on research voyages in 2018, 2019 (Eilon et al., 2022), and 2024, in addition to 1985, 1992, 1995, 2006, and 2018 near the microplate. The data not collected on our voyage was obtained through the GEBCO Grid Subsetting App. The VGG data highlights short wavelength bathymetric signals where sediments are reduced as with our area (Sandwell et al., 2014). Magnetic anomaly picks identified by Handschumacher (1976), Munschy et al. (1996), Lonsdale (2005), and Barckhausen et al. (2008) were used where possible to date features

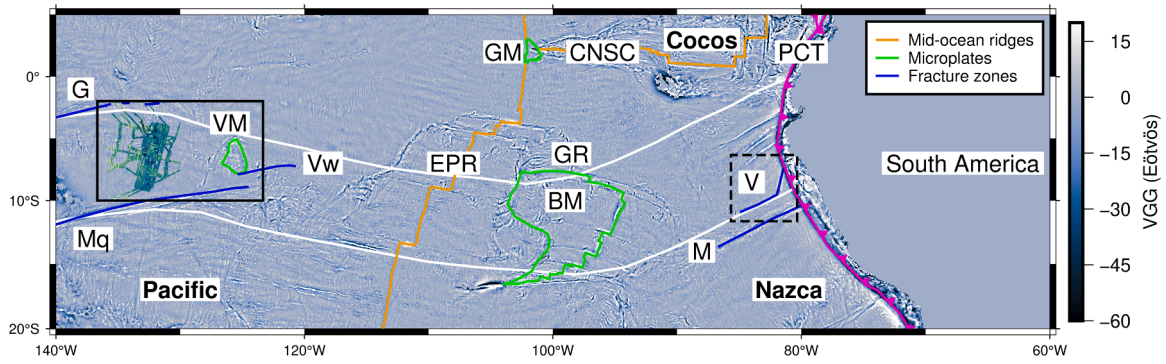


Fig. 1. SWOT VGG map (Yu et al., 2024) of the Southeastern Pacific Ocean, showing the spreading corridor of the study area. The black box is drawn around the study area, shown in Fig. 2a, and the dashed black box is drawn around the corresponding region on the Nazca Plate, shown in Fig. 3. Bathymetry in the study area is plotted in a green colorscale. Spreading ridges from Müller et al. (2019) are given by gold lines, microplates are traced in green, relevant fracture zones are denoted by blue lines, and subduction zones (Zahirovic et al., 2022) are given by pink lines. The white lines represent the outline of the spreading corridor, calculated kinematically using flowlines in GPlates 2.5.0 and the plate model form Müller et al. (2019). (VM=Viru Microplate, GM=Galapagos Microplate, BM=Bauer Microplate, Mq=Marquesas Fracture Zone, M=Mendana Fracture Zone, G=Galapagos Fracture Zone, Vw=Viru West Fracture Zone, V=Viru Fracture Zone, EPR=East Pacific Rise, CNSC=Cocos-Nazca Spreading Center, GR=Galapagos Rise, PCT=Peru Chile Trench, MqIs=Marquesas Islands).

observed in the SWOT VGG and bathymetry. Due to a lack of magnetic anomaly picks for three of the features, we could not date them directly. Instead, we estimated the ages of these features based on their distance from the available magnetic anomaly picks. We multiplied the distances from the features to nearby C20n picks with the age contrast between the nearest picks, C20n and C12n, and divided by the distance between the C20n and C12n picks. This method assumes a stagnant spreading rate for the entire period and therefore provides only a rough estimate for the ages.

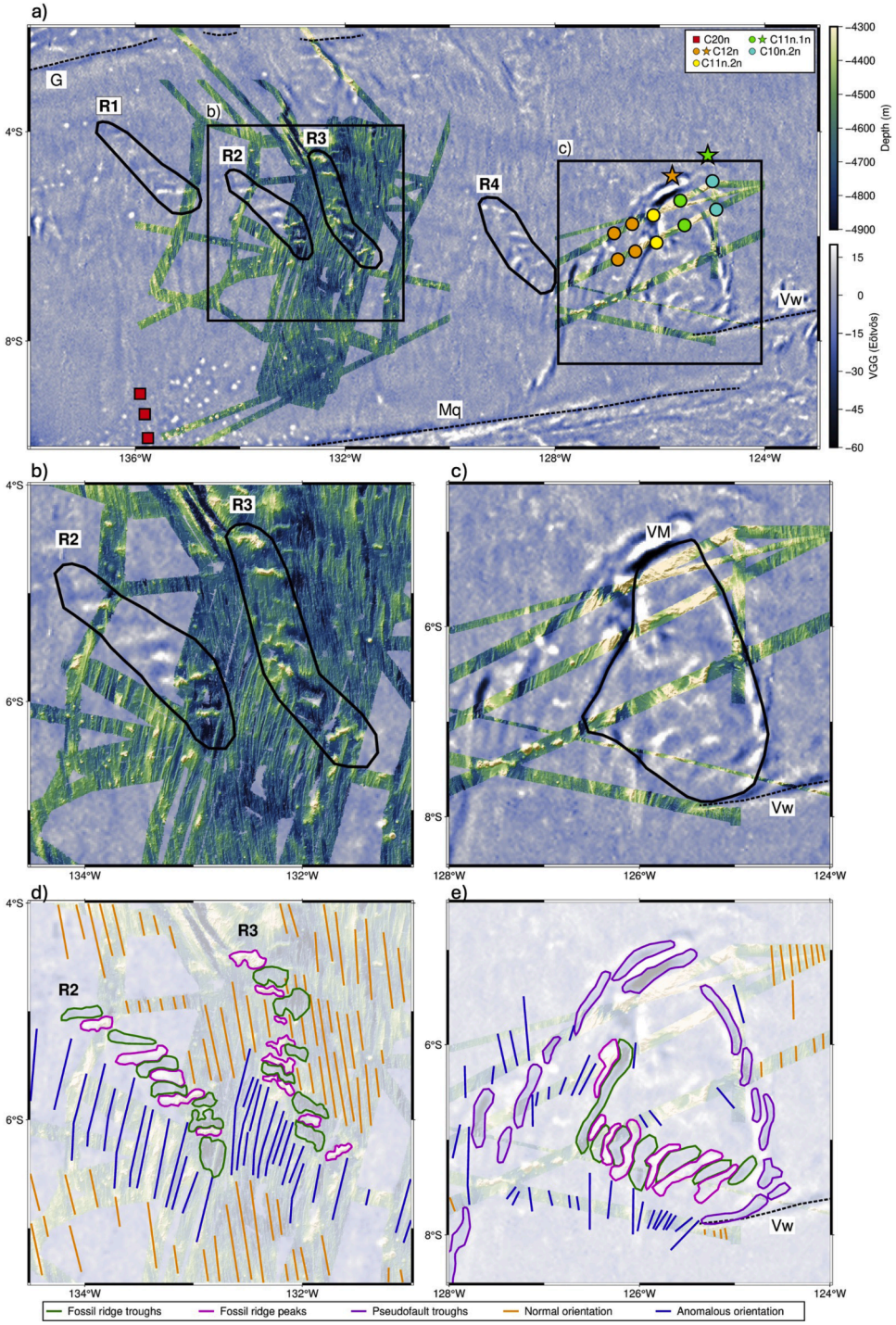


Fig. 2. a) SWOT VGG map of the study area and relevant multibeam bathymetry collected in the region. Dashed black lines represent fracture zones. Magnetic anomaly picks were color coded by chron while the shapes correspond to different datasets: squares from Munschy et al. (1996), circles from Barckhausen et al. (2008), and stars from Handschumacher (1976). Fossil ridges R1-R4 are outlined in black. Black squares are drawn around regions that correspond to panels b and c. b) Zoomed-in view of two of the fossil ridges, R2 and R3. c) Zoomed-in view of Viru microplate. Viru microplate traced in black. d) Annotations of R2 and R3 from panel b, including fossil ridge troughs and peaks in green and magenta, and the normal and anomalous abyssal hill orientations in orange and blue. e) Annotations of Viru microplate, including more fossil ridge troughs and peaks, pseudofault troughs and abyssal hill lineations. (G=Galapagos fracture zone, Vw=Viru West fracture zone, Mq=Marquesas fracture zone, VM=Viru Microplate).

2.2. Tectonic and geodynamic parameters

To assess the behavior of the mid-ocean ridge system within its broader context, we used the spreading asymmetry grid and seafloor-spreading isochrons from Seton et al. (2020), as well as three different reconstructed dynamic topography models from Müller et al. (2018a), M1, M5, and M7. Spreading asymmetry is calculated by resolving the amount of crustal accretion between neighboring seafloor-spreading isochrons and dividing the angular separation between them by the full stage rotation angle. The full calculation is provided in Müller et al. (2008). Spreading asymmetry provides the distribution of crustal accretion at a spreading ridge between two plates. If more crust is accreted on one side of the ridge, the grid assigns a percentage greater than 50 for that portion of crust, while the conjugate portion on the other flank will have a corresponding percentage <50. The value of asymmetry can provide information about the spreading conditions at the time of formation while the accompanying seafloor-spreading isochrons from Seton et al. (2020) provide bounds on each portion or "block" of crust in the asymmetry grid. Confidence grids from Seton et al. (2020) were used to evaluate the observed asymmetry values for the study region.

The dynamic topography models and their underlying plate motion models are described in Müller et al. (2018a). Models M1 and M7 do not consider mantle plumes, while M5 includes plumes. We consider model M1 at times 30 Ma and 40 Ma, and models M5 and M7 at times 29 Ma and 39 Ma. The models are available at increments of 10 Myr, and the selected times are closest to the period of interest. Mantle-driven dynamic topography can reveal anomalies that may be responsible for higher or lower topography (Parsons and Sclater, 1977), such as the South Pacific Superswell (McNutt and Fischer, 1987).

To explore large-scale plate tectonic processes, we calculated convergent velocities in GPlates 2.5.0 (Müller et al., 2018b) using the global rotation models from Müller et al. (2019) and Seton et al. (2012), for three points along the convergent margin between the Farallon and South American plates, following the methods of Wright et al. (2016). Using this method, we calculated the velocity of the Farallon plate with the South American plate anchored, for periods of 5 Myr from 55 Ma to 20 Ma at each point. This increment of 5 Myr is a condition of the underlying plate motion model (Müller et al., 2019). The coordinates of the points were: 81°W, 5°S; 76°W, 20°S; and 75°W, 35°S. The points were chosen to span the length of the spreading corridor along the South American margin (Fig. 1). Additionally, we used the kinematics tool in GPlates to extract step plot values for the convergent velocity from 50 to 20 Ma to show the change over the entire period. Müller et al. (2019) uniquely consider geologic deformation in the creation of their global plate model, a technique that had previously been used solely for regional models. This provides higher detail in their plate model than for past models such as Seton et al. (2012), specifically with the consideration of Farallon plate configuration during the early Cenozoic (Wright et al., 2016).

3. Results

3.1. Features and dates from seafloor maps

The collected bathymetry data revealed some unexpected features (Fig. 2a). Since the study area is contained within one ridge segment, the abyssal hills should all be parallel to the formative spreading ridge (Menard and Mammerickx, 1967). However, we observed two distinct abyssal hill azimuths in the data. The abyssal hills are identified as small amplitude lineations in the mapping data. The “normal” orientation group, about −10°, aligns perpendicular to the orientation of the nearby Marquesas (81°) and Galapagos (77°) fracture zones (Fig. 2a, b, d). However, the second “anomalous” orientation group, about 10°−20°, does not align perpendicularly with the nearby fracture zones, suggesting a more complex formation explanation than simple spreading and faulting (Vine and Matthews, 1963; Macdonald et al., 1996).

In addition to the differing abyssal hill orientations, the SWOT VGG and bathymetry data revealed four stair-step patterns in the seafloor, R1, R2, R3, and R4, characterized by alternating gravity and bathymetry highs and lows ranging from 30–50 km in length that step in the northwest direction (Fig. 2a, b, d). Directly to the south of these stair-step patterns, the abyssal hills are oriented at the anomalous orientation, while those directly to the north of the stair-steps are oriented in the normal direction. The anomalous abyssal hills extend from the stair-steps to the southwest by about 50–100 km, south of which the remaining abyssal hills in the ridge segment are oriented at the normal orientation. This pattern is found at each of the stair-steps but is especially apparent at R2 and R3 due to the availability of high-resolution seafloor bathymetry in the vicinity (Fig. 2b).

To the east of the four stair-steps is a complex shape, which is the previously unidentified Viru microplate. There are more stair-steps lining one of its edges extending from about 127°W, 7°S to 125°W, 8°S (Fig. 2c, e). The troughs and peaks forming this stair-step range from 30–80 km in length, with some features longer than in R1-R4 while maintaining a similar pattern of alternating highs and lows. One of the northernmost troughs is significantly longer than the rest, at around 140 km. North of this trough, the pattern is interrupted. Extending from the northern edge of the microplate to the southwest, are a series of troughs that resemble en echelon fractures. Each trough extends further south and west than the previous one. The eastern edge of the microplate also contains these en echelon troughs, with each extending further south and east than the one before, similar to patterns identified at the Mathematician microplate (Mammerickx et al., 1988). The Viru West fracture zone extends from the southern edge of the microplate. The abyssal hill lineations near the Viru microplate are more confusing than those around features R1-R4. Abyssal hills east of the microplate are close to the “normal” orientation (−4° to −9°), approximately perpendicular to the azimuth of the Viru West fracture zone (79°). The other lineations do not match the orientation perpendicular to the Viru West fracture zone or the regularity of the anomalous orientations found near R1-R4. There is also a clear pattern of abyssal hill fanning around 126°W, 8°S and 127.2°W, 7.5°S.

There are few available magnetic anomaly picks in the study area from Handschumacher (1976); Munschy et al. (1996), and Barckhausen et al. (2008). Picks for C20n coincide with the western edge of R1, while picks for C12n (30.939 and 30.479 Ma), C11n.2n (30.098 Ma), C11n.1n (29.662 and 29.401 Ma), and C10n.2n (28.745 Ma) coincide with Viru microplate. The picks for C12n and C11n.1n are found both within the microplate core and to the north, with an offset of about 130 km and 75 km, respectively, in the paleo spreading direction (Seton et al., 2020). We estimated the ages and spreading periods of R1-R4 using the method described in Section 2.1. Between the picks for C20n and C12n, there is a time span of 12.85 Ma spread over about 1034 km. We multiplied the distances between the C20n picks and the NW and SE ends for each of R2-R4 in the paleo spreading direction (Seton et al., 2020). We approximate the age ranges to be: 43.789 to 41.393 Ma for R1, 40.458 to 38.818 Ma for R2, 38.507 to 37.314 Ma for R3, and 34.133 to 32.704 for R4.

When seafloor spreads, complimentary features are formed on both sides of the ridge, creating conjugate areas of the same age on either side of the ridge (Vine and Matthews, 1963). Regions corresponding to chron C13n (33.058 Ma) and earlier between the Mendaña fracture zone and Sarmiento ridge on the Nazca plate have been subducted at the Peru-Chile trench, which should have contained complementary features to R1-R4. The remaining area contains picks for C12n (30.939 Ma) through C7n.2n (24.835 Ma), corresponding to the region on the Pacific plate containing the Viru microplate. Here, the Viru Fracture Zone, which is conjugate to the Viru West fracture zone on the Pacific plate, extends parallel to the Mendaña fracture zone until about 140 km west of the Peru-Chile trench (Fig. 3). Then, a roughly 230 km long trough extends northeast from the Viru fracture zone termination site to the Peru-Chile trench, at an orientation of around 11° from North. Magnetic

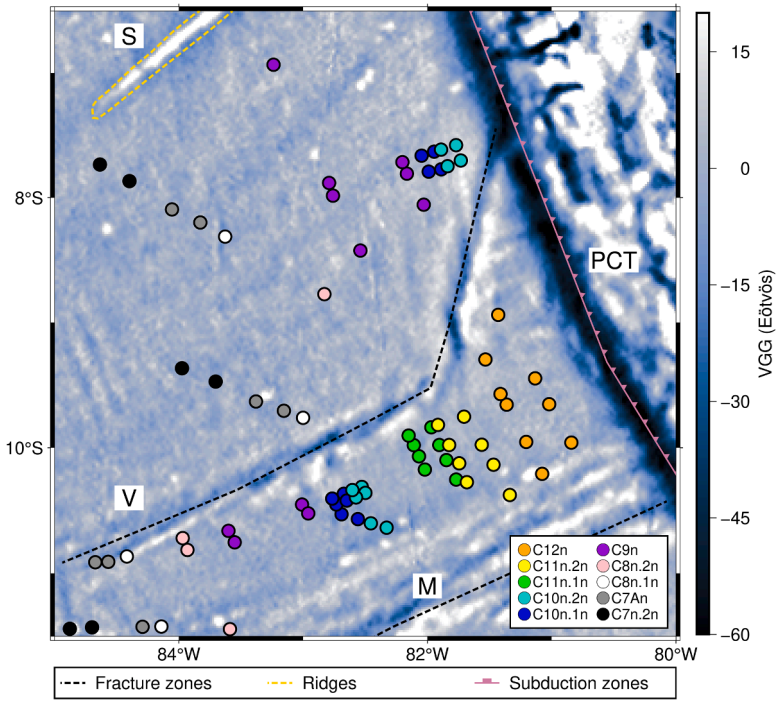


Fig. 3. SWOT VGG of the region conjugate to the study area on the Nazca plate (dashed black box in Fig. 1). Magnetic anomaly picks from Lonsdale (2005) color coded by chron. Fracture zones plotted as black dashed lines, subduction zones in pink, and ridges in yellow. (S=Sarmiento ridge, V=Viru fracture zone, M=Mendaña fracture zone, PCT=Peru-Chile trench).

anomaly picks from Lonsdale (2005) show that older crust sits adjacent to younger crust on either side of the Viru fracture zone and the 11° trough, with older crust to the south and younger crust to the north. Across the magnetic anomaly picks, there seems to be a common offset of about 200 km in the paleo spreading direction between picks of the same chron on either side of the Viru fracture zone.

3.2. Spreading asymmetry

Spreading asymmetry for the area was determined using the grid of Seton et al. (2020). The blocks of crust containing the four fossil ridges, R1-R4, range in age from 47.9 Ma to 40.1 Ma and 40.1 Ma to 33.1 Ma, according to the Seton et al. (2020) seafloor spreading isochrons. Spreading asymmetry information is unavailable for these blocks as the corresponding areas on the Nazca plate have been subducted. However, the adjacent block, corresponding to a time range of 33.1 Ma to 25.3 Ma, remains on the Nazca side. On the Pacific plate, this block contains the Viru microplate and has an asymmetry of 55–60% (Fig. 4). On the Nazca plate, this block has a corresponding asymmetry of 40–45%, indicating that more crust was accreted on the Pacific plate than the Nazca plate during this period of spreading. Based on a confidence grid of seafloor spreading parameters from Seton et al. (2020), the 47.9 to 40.1 Ma and 40.1 to 33.1 Ma blocks containing R1-R4 have a low confidence rating. Conversely, the block containing the Viru microplate corresponding to 33.1 to 25.3 Ma has a high confidence rating.

3.3. Dynamic topography

The reconstructed dynamic topography models provide information about the amount that true seafloor depth deviates from the depth predicted by depth-age relationships, which we used to determine the possible geodynamic causes for the chaotic sequence. The locations of nearby hotspots coincide with regions where the residual topography is positive, demonstrating unexpected topographic highs resulting from

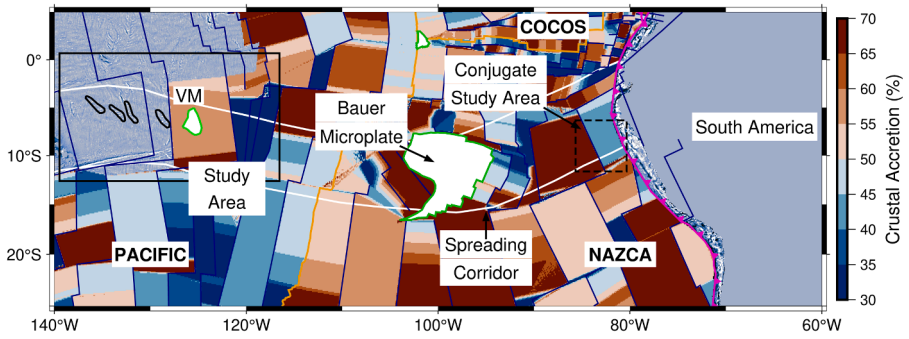


Fig. 4. Map of SWOT VGG of the spreading corridor beneath the spreading asymmetry grid from Seton et al. (2020). The black box shows the study area and the black dashed box shows the conjugate region on the Nazca flank. Relevant microplates are traced in green and spreading ridges and the subduction zone are traced in orange and magenta, respectively. Outlines of the fossil spreading ridges from Fig. 2a are plotted in black. The spreading corridor from Fig. 1 is outlined in white. Seafloor spreading isochrons from Seton et al. (2020) are plotted in dark blue. Areas of red indicate crustal accretion above 50% and areas of blue indicate crustal accretion below 50%.

hotspot magmatism and the South Pacific Superswell (Fig. 5). The positions of the fossil ridges, R1-R4, as well as Viru microplate and the Marquesas hotspot have been plotted. We have also included the perceived direction of ridge propagation, in the northeast direction. A circle of 1000 km has been plotted around the Marquesas hotspot (Whittaker et al., 2015), representing the theorized extent of hotspot influence.

The four fossil ridges lie on the edge of the 1000 km extent from the Marquesas hotspot. Additionally, across all models during the first time, 40 Ma for M1 and 39 Ma for M5 and M7, the propagation direction points down topographic gradients, from higher to lower dynamic topography. During the second time, 30 Ma for M1 and 29 Ma for M5 and M7, the propagation direction once again aligns down topographic gradients for M5 and M7. While it is not as clear for M1 at the second time, the arrow still points from higher to lower topography.

3.4. Convergence velocity

To assess whether the period of chaotic spreading corresponded to changes in the motion of adjacent plates, we examined the shift in convergence rate and direction along the South American margin. The convergence velocity of three points on the Farallon plate with respect to the South American plate were calculated using GPlates 2.5.0 velocity calculations, following the methodology of Wright et al. (2016). This was done for two plate models, Müller et al. (2019) and Seton et al. (2012), referred to here as M19 and S12.

For M19, before the ridge propagation events that created R1-R4, the convergence velocity from 50 to 45 Ma was 88 mm/yr at point A, 59 mm/yr at point B, and 26 mm/yr at point C, represented by the blue arrows in Fig. 6a. Around the formation time of the microplate, the convergence velocity from 30 to 25 Ma was 130 mm/yr at point A, 119 mm/yr at point B, and 102 mm/yr at point C, showing an increase in velocity at all three points over the course of the chaotic spreading event. This change is reflected in the longer length of the red arrows compared to the blue arrows, representing an increase in the strength of convergence between the Farallon and South America plates over the course of the period of interest (Fig. 6a). The maximum rate for point A was reached during the period of 40 to 35 Ma, at a rate of 142 mm/yr. For points B and C, the maximum rate occurred during the period of 30 to 25 Ma at the rates listed above, near the end of the chaotic spreading sequence. During the period of 50 to 45 Ma, the obliquity was 86° at point A, 93° at point B, and 100° at point C. In contrast, during the period from 30 to 25 Ma, the obliquity was 76° at point A, 75° at point B, and 72° at point C.

For S12, the convergence velocity during the period from 50 to 45 Ma, represented by a dashed blue arrow in Fig. 6a, was 121 mm/yr at point A, 90 mm/yr at point B, and 54 mm/yr at point C. During the time that Viru microplate formed, 30 to 25 Ma and represented by a dashed red arrow, the convergence velocity was 109 mm/yr at point A, 97 mm/yr at point B, and 81 mm/yr at point C, demonstrating an increase at points B and C, but not at point A. These changes are similarly reflected in the relative lengths of the arrows. The maximum rates for the three points during the period of interest (44 Ma to 28 Ma) were 116 mm/yr at point A during 45 to 40 Ma, and 97 mm/yr at point B and 81 mm/yr at point C during 30 to 25 Ma. During the 50 to 45 Ma, the obliquity was 87° at point A, 92° at point B, and 94° at point C. From 30 to 25 Ma, the obliquity was 70° at point A, 69° at point B, and 65° at point C.

Between the two models, the convergence velocity increases from the 50 Ma to 45 Ma period to the 30 Ma to 25 Ma period, except at point A in model S12. Here, the convergence velocity was greater during the former period. There are also larger contrasts between the two times for model M19. Across all points and models, the obliquity of the arrows decreases from the earlier to the later period, indicating a northward change in direction of the convergence velocity of the Farallon plate.

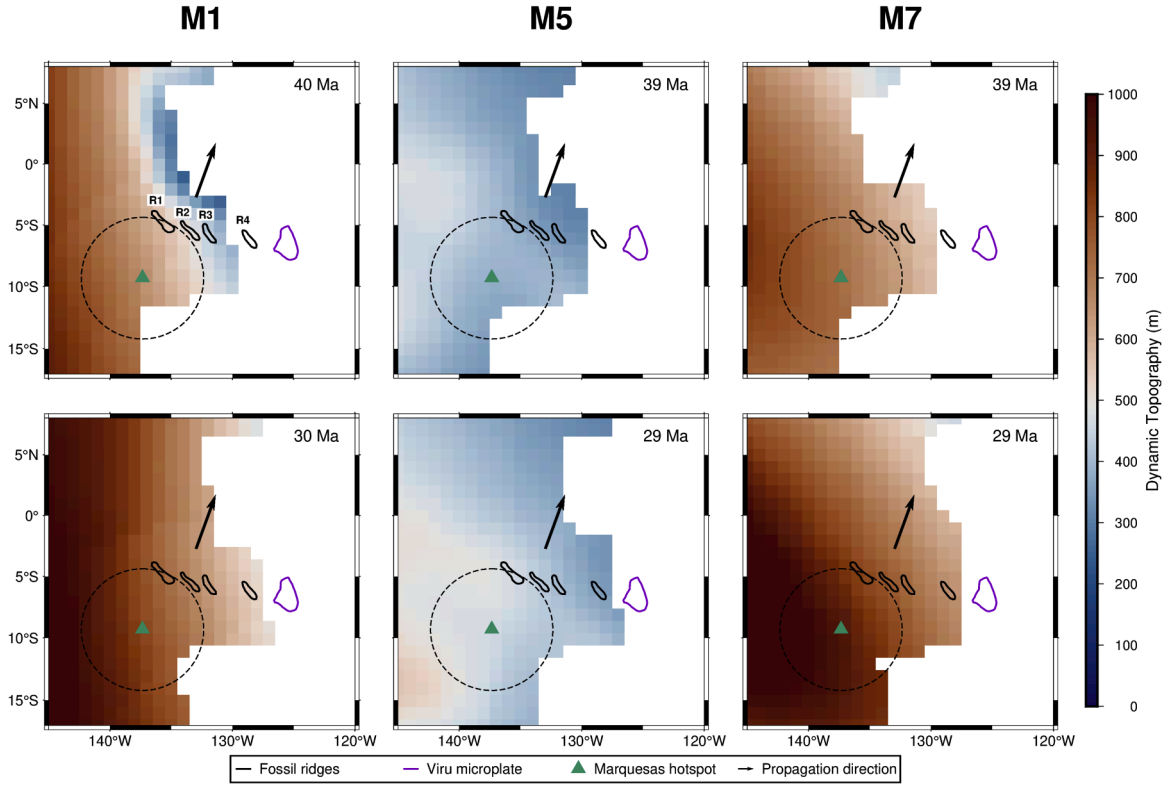


Fig. 5. Reconstructed dynamic topography models M1, M5, and M7 from Müller et al. (2018a) at times 40 and 30 Ma for M1 and 39 and 29 Ma for M5, M7. The Marquesas hotspot is plotted as a green triangle, and its 1000 km extent is represented by the dashed circle. The four fossil ridges, R1-R4 are outlined in black, and the Viru Microplate is outlined in purple. The black arrow shows the propagation direction interpreted from the relative orientations of abyssal hills, highlighting that the gradient of topography in the propagation direction is consistently downslope.

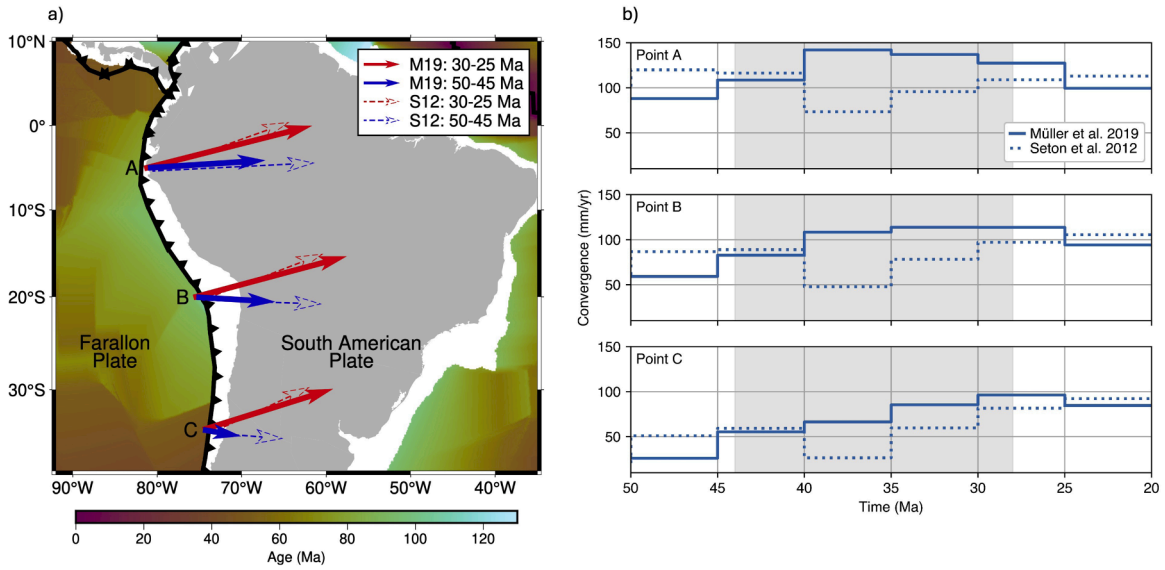


Fig. 6. a) Convergence of the Farallon Plate with respect to the South American Plate, based on Fig. 34 in Wright et al. (2016), for two different plate models, Müller et al. (2019) (M19) and Seton et al. (2012) (S12). Arrows represent the relative magnitude and azimuth of the convergence during the periods from 30 to 25 Ma (solid red arrow for M19, dashed red arrow for S12) and 50 to 45 Ma (solid blue arrow for M19, dashed blue arrow for S12). The seafloor age grid at 30 Ma and the reconstructed 30 Ma coastlines, both from Müller et al. (2019), are plotted in the background. Plate boundaries from Zahirovic et al. (2022) are plotted in black. The arrows corresponding to the later period are larger in magnitude and have a more northward azimuth, indicating a change in velocity from before the chaotic period to after. b) Step plots of the convergence velocity at each of the three points for both models. Solid blue lines for M19 and dashed blue lines for S12. The gray rectangle shows the timing of the chaotic spreading sequence. Values were calculated using the GPlates 2.5.0 kinematics tool using the two different plate models.

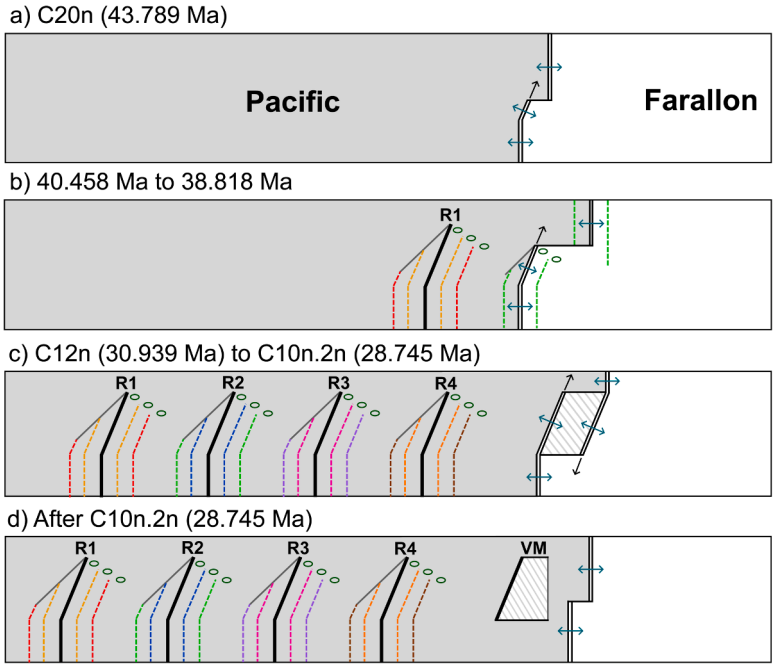


Fig. 7. The proposed sequence of events for the chaotic spreading period, 43.789 Ma to 28.745 Ma, at the ridge segment between the Galapagos and Marquesas fracture zones. Grey represents the Pacific Plate and white represents the Farallon Plate. The striped grey region represents Viru microplate, labeled VM. The dark turquoise arrows show spreading directions and the double lines represent active spreading ridges. The dark grey lines represent pseudofaults. Fossil ridge troughs are drawn in dark green along the inner pseudofaults. Thick black lines represent fossil ridges. The dashed colored lines represent arbitrary isochrons and tectonic fabric azimuths for each ridge. Red and yellow are times 1 and 2 corresponding to R1, green and blue are times 3 and 4 corresponding to R2, purple and pink are times 5 and 6 corresponding to R3, and finally brown and orange are times 7 and 8 corresponding to R4. (a) Spreading and ridge propagation began at C20n (43.789 Ma). The first ridge jump occurred around 41.393 Ma creating the first fossil ridge, R1. (b) Spreading resumes at a ridge to the east around 40.458 Ma until 38.818 Ma. The same process is repeated for the remaining fossil ridges. The propagations and jumps for R2, R3, and R4 are therefore omitted. (c) After the four fossil ridges formed, the Viru Microplate formed between two spreading ridges starting around C12n (30.939 Ma). (d) The Viru Microplate (VM) detached near C10n.2n (28.745 Ma) with the death of the western ridge, while spreading resumes at the eastern ridge. The microplate is defined between the western fossil ridge and the eastern inner pseudofault.

4. Discussion

4.1. Mapping interpretation

The stair-step features (R1-R4) and microplate margins in the study region resemble abandoned spreading ridge and pseudofault traces resulting from propagating ridges (Hey, 1977) (Fig. 2d). The repeated troughs and peaks suggest a series of small ridge jumps (Hey, 1977; Mammerickx et al., 1988). These similarities indicate that R1-R4 are four fossil ridges formed due to ridge propagation. Additionally, the distinct abyssal hill orientations form oblique tectonic fabric, which is a key indicator of propagating ridges (Hey, 2021). Based on the observed geometry of the oblique abyssal hill fabric, we propose a northeastward propagation with a heading of about 10°−20° (Fig. 7). Additionally, the existence of separate fossil ridges indicates that each has been abandoned at some point in favor of a new spreading ridge, likely resulting from larger ridge jumps.

We interpret the Viru microplate as a lithospheric block formed between two spreading ridges which we refer to here as the western and eastern ridges of the microplate. As each ridge propagated, the western and eastern inner and outer pseudofaults formed. The southwestern edge consists of fossil ridge formations similar to R1-R4, indicating that this edge is one of the bounding spreading ridges that created the microplate. We suggest that the western series of troughs extending toward the Marquesas fracture zone is the outer pseudofault of the western abandoned spreading ridge. The eastern series of troughs extending southeast marks the inner pseudofault of the eastern ridge and represents where full spreading resumed after the western axis failed, which would have been transferred to the Pacific plate along with the abandoned western spreading ridge. We define the Viru microplate as the piece of lithosphere contained between the western fossil spreading ridge and the eastern inner pseudofault (Fig. 2e). The eastern inner pseudofault terminates at the Viru West fracture zone, suggesting that some transform behavior occurred at the eastern ridge, following the abandonment of the western ridge, also creating the complementary Viru fracture zone on the Nazca plate (Fig. 3). The offset of identified magnetic anomaly picks on either side of the western outer pseudofault demonstrates the propagation of the western ridge during microplate formation, with younger crust lying adjacent to older crust (Hey, 1977). Based on Fig. 3 from Hey (1977), we expect the crust north of the western inner pseudofault to be east of the matching age to the south, matching with the relative positions of the magnetic anomaly picks in our study region (Fig. 2a).

On the conjugate flank, we suggest that the apparent change in direction of the Viru fracture zone is not the fracture zone itself, but the eastern outer pseudofault formed by the eastern spreading ridge of the Viru microplate. This is supported by the 200 km offset of matching magnetic anomalies on either side of the pseudofault. The outer pseudofault of the remaining spreading ridge would have remained on the Farallon/Nazca side following the cessation of spreading at the western spreading ridge. Huchon and Bourgois (1990) similarly mentioned that the trough sitting at 11° may be an outer pseudofault resulting from ridge propagation, while Lonsdale (2005) interpreted this section as a failed rifting site, showing some agreement with our analysis. Following the cessation of spreading at the western ridge, if normal spreading resumed at the eastern ridge, then the eastern inner and outer pseudofaults would have been separated on either side of the ridge, with some transform behavior occurring to create the Viru and Viru West fracture zones.

4.2. Chaotic sequence of events

We infer that four northeast-propagating ridge episodes occurred between C20n (43.789 Ma) and 32.704 Ma, estimated from the distance between R4 and picks for C20n, which created features R1-R4. Each propagating episode involved short-term propagation and eventual abandonment of a spreading center, creating the observed stairstep pattern of alternating troughs and peaks captured by the Pacific plate after each ridge jump. The episodes lasted during the estimated periods provided in Section 3.1. Since abyssal hills typically form parallel to the formative spreading ridge, the seafloor created by the northeast-oriented propagating ridge segment would have created the anomalous abyssal hills observed in the data (Fig. 7). The study area is near the edge of the Marquesas hotspot's 1000 km extent. However, if it had caused the ridge propagation and jumps observed, then they would have advanced towards the hotspot rather than opposite in the northeast direction. Instead, we suggest that ridge propagation and jumping advanced down topographic gradients in the northeast direction, as evidenced by the downhill gradients observed in the three dynamic topography models and orientations of lineations in the area (Fig. 5). The relative proximity of the R1-R4 suggests shorter intervals of time between each episode.

The larger distance from R4 to the Viru microplate indicates a brief return to non-complex spreading before dual spreading began at the Pacific-Farallon plate boundary around C12n (30.939 Ma), initiating the formation of the Viru microplate. The western ridge propagated northeast, creating the observed western outer pseudofault, while the eastern ridge propagated southwest, creating the eastern inner and outer pseudofaults. As propagation and spreading advanced, the pseudofaults, the stair-step patterns, and the core of the microplate grew. Spreading at the western ridge likely ceased around C10n.2n (28.745). As a result, the western outer pseudofault and the western fossil ridge were transferred to the Pacific plate, forming the fossil ridge pattern and en echelon troughs on the western edge of the microplate. Propagation ceased at the eastern ridge while normal spreading continued. The eastern inner pseudofault was also transferred to the Pacific plate, forming the eastern edge of the microplate. The eastern outer pseudofault remained on the Farallon plate and can now be identified as the 230 km trough oriented at 11° connecting the Viru fracture zone to the Peru-Chile trench. Transform behavior occurred at the spreading ridge, creating the Viru West and Viru fracture zones leading from the microplate and the eastern outer pseudofaults. At this point, the orientation of the ridge returned to the normal orientation, creating the more regular abyssal hill lineations that we observe east of Viru microplate.

Seafloor-spreading isochrons from Seton et al. (2020) suggest that R2-R3 formed between 44 Ma and 40.1 Ma, while R4 formed during 40.1 Ma and 33.1 Ma (Fig. 4). However, our calculations based upon spacing of the magnetic anomalies and the fossil ridges suggest that R3 formed by 37 Ma, extending the duration of the 44–40.1 Ma block. Seton et al. (2020) assign an unexpectedly large width of crust for the 44–40.1 Ma block and a surprisingly smaller width for the 40.1–33.1 Ma block, even though it spans a longer time duration. Since the underlying plate model is based upon magnetic anomaly picks (Seton et al., 2020; Müller et al., 2019), and there are no magnetic anomaly picks here, it is unclear why the 44–40.1 Ma block has a larger width of crust. We suggest that there are issues with the calculations of seafloor spreading isochrons in this area, supported by the low confidence rating of the spreading parameters for the blocks containing R1-R4 (Seton et al., 2020). This will be difficult to improve upon without the collection of additional magnetic anomalies. The 33.1 Ma to 25.3 Ma block containing Viru microplate was assigned a higher spreading asymmetry of 55–60%, and has a high confidence rating (Seton et al., 2020), supporting our interpretation that the Viru microplate was transferred to the Pacific plate and creating the observed spreading imbalance.

4.3. Tectonic sequence of events

The convergence calculations show a change in velocity of the Farallon plate with respect to the South American plate around the time of the chaotic spreading period. This indicates that the spreading sequence may be related to the changes in the broader stress regimes of the tectonic plates at the time. Here we consider some major events that

occurred from 53 Ma to 21 Ma pertaining to the Pacific and Farallon plates (Fig. 8).

Around 52–45 Ma, the Pacific Plate changed its direction of motion by about 30 degrees in the counterclockwise direction and increased in speed, contributing to the bend in the Hawaiian-Emperor Seamount Chain along with southward drift of the Hawaiian hotspot (Gaastra et al., 2022; Torsvik et al., 2017; Barckhausen et al., 2013; Menard and Atwater, 1968). Typical explanations for this change are a major plate reorganization related to the initiation of subduction at the Izu-Bonin-Mariana and Tonga-Kermadec subduction zones around 52 Ma (Arculus et al., 2015; Meffre et al., 2012; Sutherland et al., 2020). The initiation of subduction at the Izu-Bonin-Mariana margins may have been caused by the initiation of the India-Eurasia collision lasting from 52 to 49 Ma (Gaastra et al., 2022; Bouilhol et al., 2013). The change in Pacific plate motion only a few million years before the first ridge propagation event around C20n (43.789 Ma) suggests that the re-organizations leading to the shift may have influenced the processes at the Pacific-Farallon spreading ridge. The increased pull to the west on the Pacific plate altered the stress regime at its boundaries, leading to instability at the ridge segment and initiating the complex spreading that contributed to the observed spreading asymmetry and unbalanced crustal accretion between the Pacific and Farallon plates at this time in our study region.

The breakup of the Farallon plate initiated around 28 Ma (Atwater and Stock, 1998), although a portion of it known as the Vancouver plate had broken off earlier around 52.6 Ma (Atwater and Winterer, 1989). As the Pacific-Farallon spreading ridge approached the North American subduction zone, the section of the Farallon plate between the North American plate and the Pacific plate broke into several smaller plates: the Monterey and Arguello plates between the Pioneer and Murray fracture zones (Atwater, 1970; Atwater and Winterer, 1989; Lonsdale, 2005) as well as the Jasper plate between the Murray and Molokai fracture zones, around 28.5 to 27.5 Ma (Lonsdale, 2005). This split contributed to the loss along the Farallon plate's North American subduction margin, such that the remaining slab pull on the northern portion of the plate came primarily from subduction along the Middle America trench. The more westerly strike of this portion resulted in an increased northward slab pull on the northern portion of the plate (Lonsdale, 2005). This created a divergence of slab pull forces on the Farallon plate from the Middle America trench and the South American margin, leading to the split that created the Cocos and Nazca plates around 23 Ma (Lonsdale, 2005).

The observed increased convergence velocity and northward azimuth shift from 50–45 Ma to 30–25 Ma likely reflects this loss of the northern Farallon subduction margin and the resulting northward slab pull on the Farallon plate (Fig. 6a). While both the S12 and M19 models show a general increase in convergence rate, model S12 predicts a decrease in rate at point A. However, S12 does predict an increase from 40–35 Ma to 30–25 Ma (Fig. 6b). While the two plate models provide similar results, the inclusion of finer detail and deformation in the M19 model likely contributes to the observed difference at point A. The general similarities between the two models impart high confidence in the results at all three points for the direction and points B and C for the rate, while we are inclined to agree with the result for rate at point A for M19 since it incorporates higher complexity at the Farallon plate's boundaries during this time. These shifting forces contributed to the breakup of the Farallon plate and spreading regime changes at its boundaries, leading to the capture of the Viru microplate around C10n.2n (28.745 Ma).

Plate reorganizations occur on both global and regional scales, with global events creating significantly longer instability periods of up to 20–25 Myr, compared to only 2 Myr for regional changes (Janin et al., 2025). Additionally, major changes that occurred in the western Pacific region that led to its change in plate motion was likely a global plate reorganization event (Janin et al., 2025). The major change in Pacific plate motion around 52–45 Ma (Gaastra et al., 2022) likely caused instability along plate boundaries globally for the following 20–25 Myr, until around 32 to 27 Ma. The formation of features R1-R4 and the Viru microplate during C20n (43.789 Ma) to C10n.2n (28.745 Ma) falls within this 20–25 Myr period of instability (Fig. 8). We suggest that the complex spreading behavior observed occurred as a result of this Pacific global reorganization. Furthermore, the cessation of spreading at the Viru microplate's western axis likely marks the restoration of plate stability and a return to more stable spreading configuration (McKenzie and Morgan, 1969), in addition to spreading regime changes resulting from the breakup of the Farallon plate. Confirmation of this link between widespread plate instability and localized complex spreading behavior requires further exploration of seafloor formed at this time across the

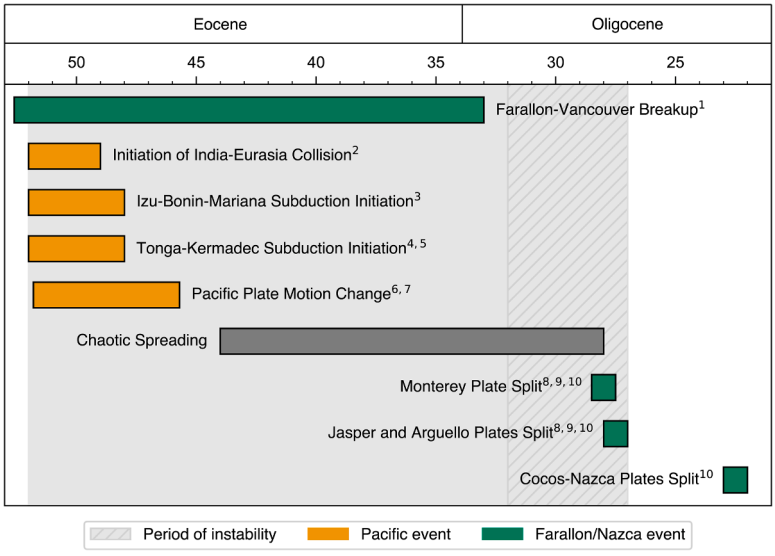


Fig. 8. Table of key tectonic events from 53 to 21 Ma. The events are categorized as Pacific, and Farallon/Nazca events. The proposed period of instability from Janin et al. (2025) is plotted in the background as the light gray rectangle spanning 47 to 27 Ma. The diagonal gray lines show the possible range for this period of instability (up to 22 Ma). The change in Pacific Plate motion direction predates the chaotic spreading sequence, while the breakup of the Farallon Pate coincides with the ending of the chaotic spreading sequence. 1 - Atwater and Winterer (1989); 2 - Bouilhol et al. (2013); 3 - Arculus et al. (2015); 4 - Meffre et al. (2012); 5 - Sutherland et al. (2020); 6 - Barckhausen et al. (2013); 7 - Menard and Atwater (1968); 8 - Atwater (1970); 9 - Atwater and Winterer (1989); 10 - Lonsdale (2005).

globe.

5. Conclusion

The identification of unexpected abyssal hill patterns, fossil ridges, and a previously undiscovered microplate between the Marquesas and Galapagos fracture zones on the Pacific plate led to the recognition of a chaotic spreading sequence in the crust formed between C20n (43.789 Ma) to C10n.2n (28.745 Ma). The formation of these features coincides with the timing of large-scale tectonic events which led to changes in the stress regime of the Pacific and Farallon plates. It is therefore postulated that these events, the change in Pacific plate motion direction and the breakup of the Farallon plate, may have contributed to the beginning and end of the chaotic spreading sequence that created the four fossil ridges and the Viru microplate. The features are believed to have been formed from a combination of ridge propagation and jumping, as evidenced by the alternating peaks and troughs, en echelon pseudofault traces, and oblique abyssal hill orientations. Overall, we find that large-scale tectonic events that affect the stress regime of the tectonic plates could cause periods of complex spreading at divergent boundaries, leading to plate reorganizations and the confusing features characteristic of the Southeast Pacific Ocean basin.

CRediT authorship contribution statement

Josephine K. Joergensen: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft. **Maria Seton:** Funding acquisition, Methodology, Resources, Software, Supervision, Validation, Writing – review & editing. **R. Dietmar Müller:** Methodology, Resources, Software, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors have no conflicts of interest to declare.

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Data availability

A link to the data repository has been shared. Other data sources have been referenced.

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